

GNN For Neutrino Decay Classification

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Project Summary and Goals

- Solidify understanding of electron decay and the three-flavour paradigm
- Understand the goals of DUNE, and aspects used by NuGraph
- Analyze and interpret existing NuGraph codebase
- Identify shortcomings of the model and implement remedies

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- 2 Deep Underground Neutrino Experiment (DUNE)
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Neutrinos

- A neutrino is a fermion - a spin- $\frac{1}{2}$ elementary particle
- Interact with matter exclusively via weak nuclear and gravitational forces
- Minimal gravitational interaction due to their extremely small rest mass
- Weak interactions create neutrinos which fall into one of three leptonic flavours:



electron
neutrino



muon
neutrino



tau
neutrino

Neutrino Oscillations

- As a neutrino of a particular propagates through space, it can change its flavour.
- The PMNS matrix describes the mixing between different flavor eigenstates of neutrinos and their mass eigenstates:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \underbrace{\begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix}}_{\text{PMNS}} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} \quad (1)$$

Decomposing PMNS, we can compute the oscillation probability of $\nu_\mu \rightarrow \nu_\tau$,

$$P(\nu_\mu \rightarrow \nu_\tau) = \sin^2(2\theta_{23}) \sin^2 \left(\frac{\Delta m_{23}^2 L}{4E} \right) \quad (2)$$

- **Artificial Neutrinos**

- **Accelerator neutrino**
- Nuclear reactor neutrinos
- Collider neutrinos

- **Atmospheric Neutrinos**

- Result of interaction of cosmic rays and atomic nuclei in Earth's atmosphere.
- Creates showers of particles which produce neutrinos once decayed

- **Supernovae Neutinos**

- **Solar Neutrinos**

- Produced together with natural background radiation.
- Decay chains of Uranium-238 and Thorium-232 isotopes.

- **Geological Neutrinos**

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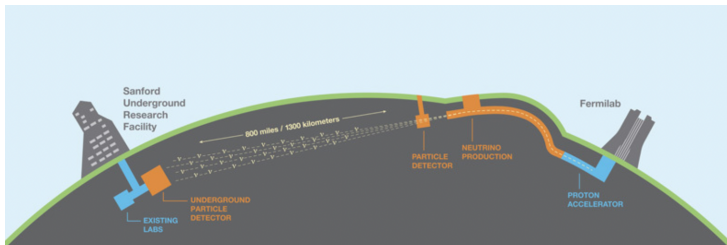
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Deep Underground Neutrino Experiment (DUNE)

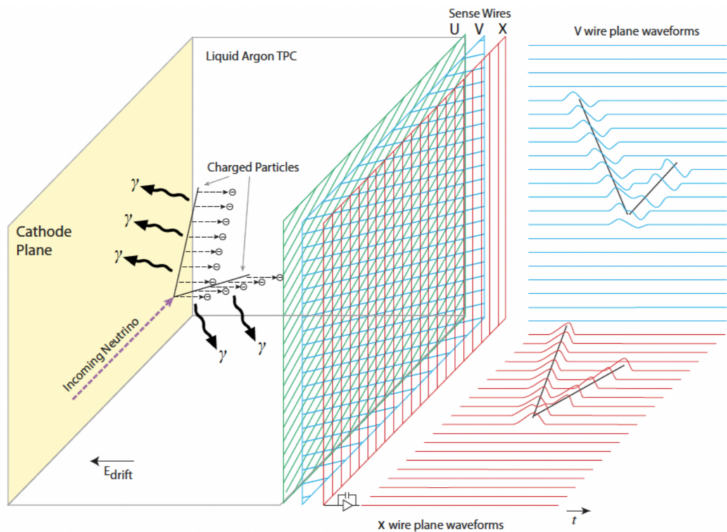
- DUNE Physics Goals: Explore neutrino oscillations, probe grand unified theories, and refine standard models with precise measurements.
- Muon Neutrino Beam: DUNE is constructing a neutrino beam to generate muon neutrinos, which will oscillate through a tunnel, providing valuable data for neutrino research.
- Supernova Preparedness: DUNE aims to be ready to detect neutrino bursts from core-collapse supernovae, necessitating continuous detector operation.
- Rare Supernova Events: Due to their rarity, at least one DUNE detector must be active at all times to capture data from potential bursts.

Deep Underground Neutrino Experiment (DUNE)

- DUNE aims to measure oscillation probabilities for muon neutrinos or antineutrinos to remain the same flavor or oscillate into electron flavor counterparts.
- Comprised of three crucial components:
 - High-intensity neutrino beam from proton accelerator.
 - Near detector near neutrino source.
 - Far detector placed at 1300km baseline, 1.5km underground.
- Far detector site: Four liquid argon time projection chamber modules for high-resolution data collection.



Time Projection Chamber



Data Collection: Time Projection Chamber

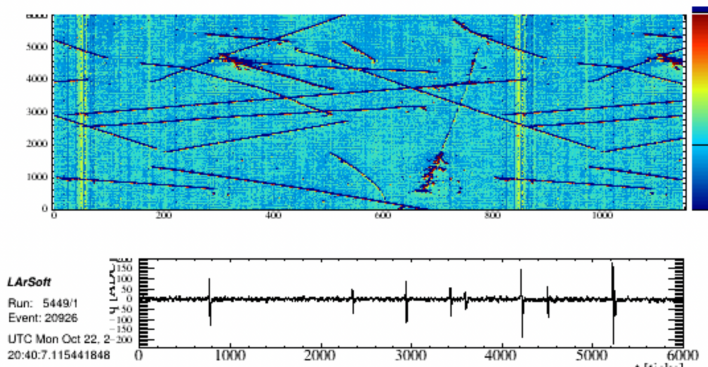


Figure: Example of pedestal-subtracted data for one ProtoDUNE-SP wire plane.
Source: [Introduction to DUNE. Vol 1](#)

Data Collection: Time Projection Chamber

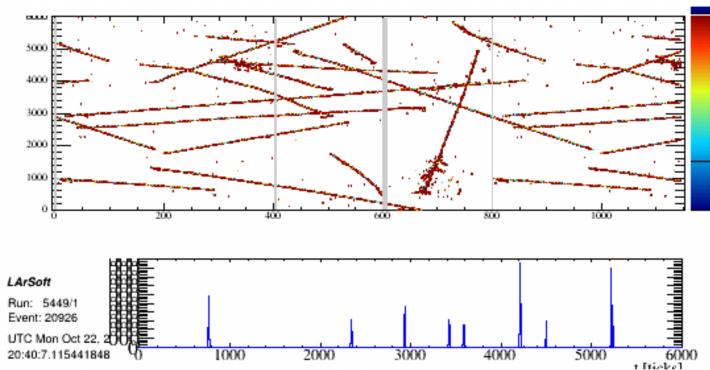


Figure: Calibrated, deconvoluted pedestal-subtracted data for one ProtoDUNE-SP wire plane. Source: [Introduction to DUNE. Vol 1](#)

Data Collection: SAMPLE.HDF5

Each data sample collected in the LArTPC is processed and stored into an hdf5 file.

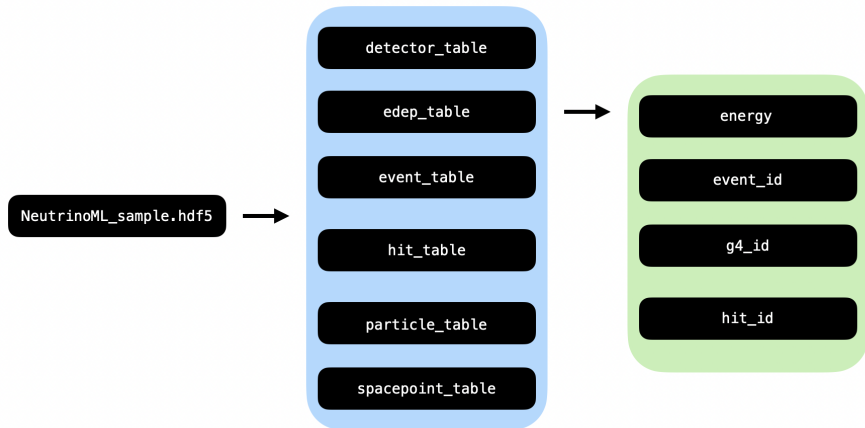


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Codebase Dependencies

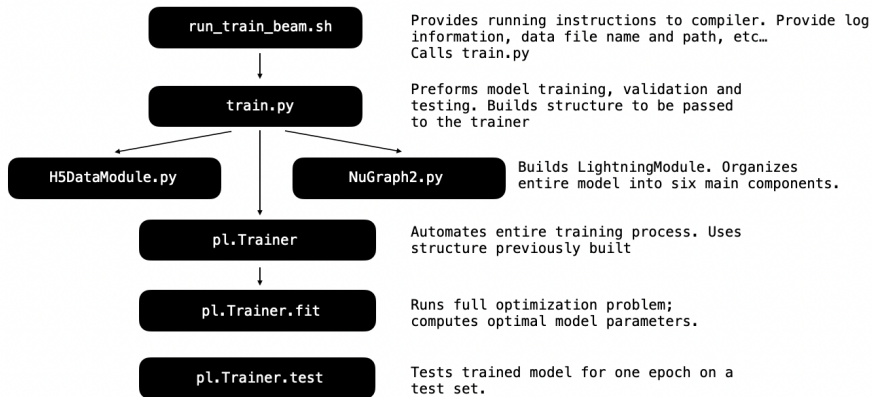


Figure: Description of NuGraph Neural Network. See [LightningModule](#) for further documentation.

NuGraph2.py Structure

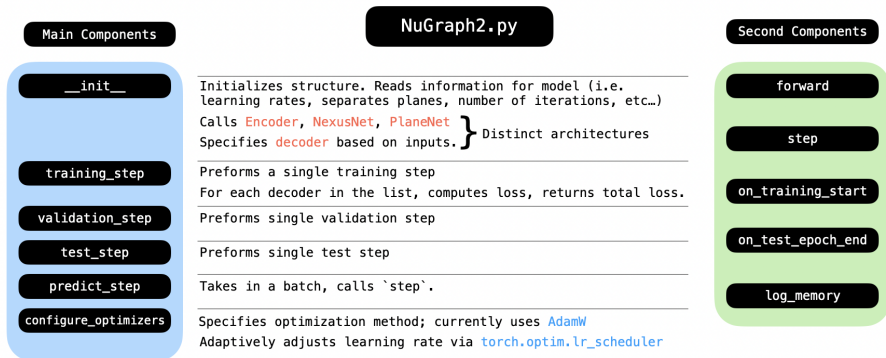


Figure: See [AdamW](#), [OneCycleLR](#).

Layer Description

Feedforward Layers

Encoder

Two-layer sequential network with one linear and one non-linear layer:

$$z = \tanh(Wx + b)$$

W, b are learned parameters

NexusNet

EdgeNet: Two-layer sequential network with Softmax.

$$z = \sigma(Wx + b) \quad (\sigma(x))_i = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

NodeNet: Four-layer sequential network.

$$z = \tanh(W_2 \tanh(W_1 x + b_1) + b_2)$$

W_i, b_i are learned parameters

PlaneNet

Very similar to NexusNet

Decoder Descriptions

DecoderBase

Base class for all NuGraph decoders
Specifies loss update based on cooling parameters
Updates metrics (loss information over time)

SemanticDecoder

Convolve down to a single score per semantic class for each plane
Uses **RecallLoss**

FilterDecoder

Convolve down to a single score, filters out noise. Uses **BCELoss**

EventDecoder

Convolve graph node features down to a single score for entire event
Uses **RecallLoss**

VertexDecoder

Uses **LogCoshLoss**

score: model's guess

Loss Functions

RecallLoss

Really just cross-entropy loss:

$$\mathcal{L}(\theta) = \text{Avg}_i CE(y_i, \sigma(f(x_i, \theta)))$$

$$CE(p, \hat{p}) = - \sum_i p_i \ln(\hat{p}_i)$$

LogCoshLoss

Uses the soft plus function;

$$X = \sqrt{\sum_i (f(x_i, \theta) - y_i)^2}$$

$$L(x) = \log(1 + \exp(x))$$

Smooth analog of ReLU

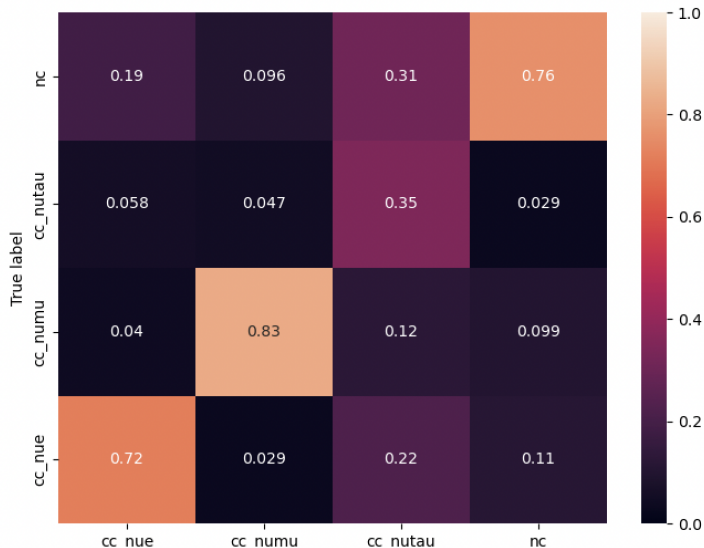
$$\mathcal{L}(\theta) = \text{Avg}_i (X + F(-2X) - \log(2))$$

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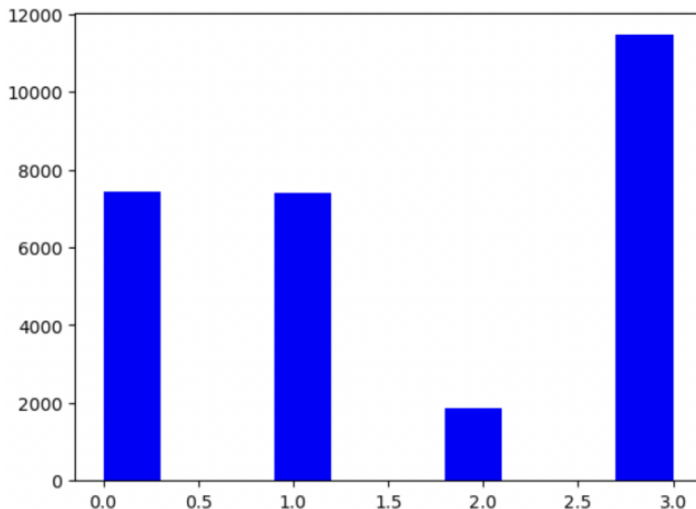
Imbalanced Dataset

Bellow is the Precision Matrix after being completely trained.



Imbalanced Dataset

The frequency of each event classification



Weighted Cross Entropy

We implemented a *modified* weighted cross-entropy loss as to combat the imbalanced data.

RecallLoss

Really just cross-entropy loss:

$$\mathcal{L}(\theta) = \text{Avg}_i \left(w_{freq}[y_i] \text{CE}(y_i, \sigma(f(x_i, \theta))) \right) \quad \text{CE}(p, \hat{p}) = - \sum_i p_i \ln(\hat{p}_i)$$

$w_{freq}[y_i]$ = relative frequency of class y_i

Encoder Depth

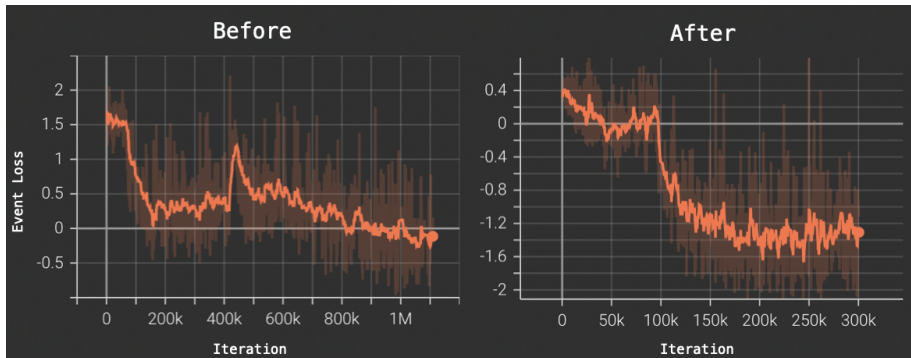
Encoder

Two-layer sequential network with alternating linear layers and nonlinear tanh activation function.

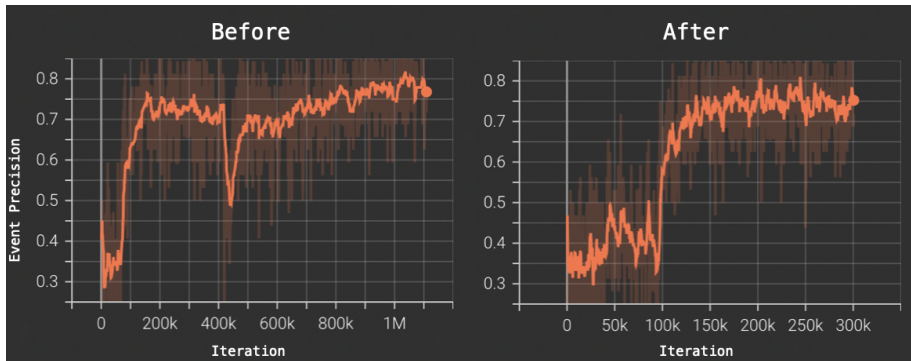
$$z = \tanh(W_2 \tanh(W_1 x + b_1) + b_2)$$

W_1, W_2 are trained weights and b_1, b_2 trained biases

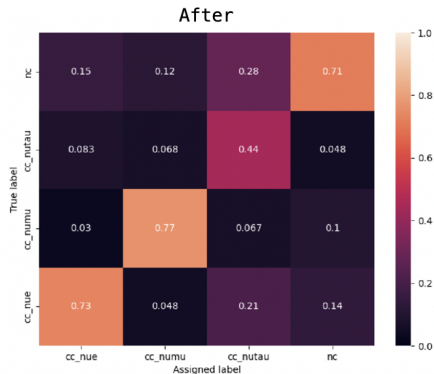
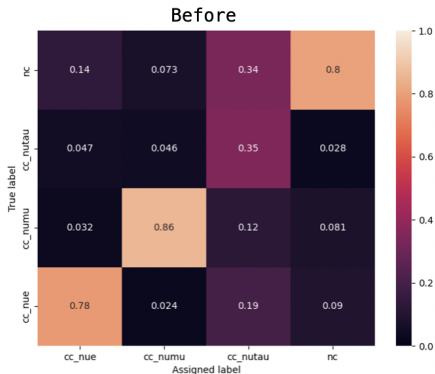
Results: Training Status



Results: Training Status



Results: Training Status



Results: Temperature Data (Extra)

